



DATA LAKEHOUSE WITH SNOWFLAKE

Big Data Engineering | Spring 2025

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1. Introduction

The project Data Lakehouse with Snowflake was designed to demonstrate how modern cloud data platforms can manage and analyse large-scale, heterogeneous datasets. The motivation for this project arises from the growing need to integrate data lakes and data warehouses into a single platform. Traditionally, data lakes were used for flexible storage of raw data, while data warehouses were used for structured analytics. However, maintaining both architectures in parallel often leads to high costs, duplication, and inefficiencies.

A Data Lakehouse solves this problem by combining the low-cost storage and flexibility of a lake with the structured governance and high performance of a warehouse. Snowflake has emerged as a leading Lakehouse platform due to its separation of storage and compute, native support for structured and semi-structured data, and ability to scale elastically in the cloud.

For this project, we used YouTube Trending Data (CSV and JSON formats) as a real-world test case. The dataset is stored on Azure Blob Storage and ingested into Snowflake, where it is cleaned, transformed, and analysed. The outcome is a consolidated analytical dataset (`table_youtube_final`) that can be directly consumed by BI dashboards and machine learning pipelines.

1.1. Role

The role of this project was to:

- Build an **end-to-end pipeline** using Azure and Snowflake.
- Securely connect Snowflake to Blob Storage using external stages.
- Load multiple file formats (CSV and JSON) into staging tables.
- Apply data cleaning and transformations to ensure accuracy.
- Integrate different data sources into a **final consolidated table**.
- Run SQL queries to answer analytical questions and identify data quality issues.
- Document the architecture, setup, challenges, and solutions.

This project serves as both a **learning exercise** in cloud-based data engineering and a **prototype** for how enterprises can adopt a Lakehouse approach for streaming or batch analytical workloads.

2. Setup

The environment setup included both **cloud configuration** and **Snowflake schema design**.

1. Data Source Setup

- Datasets provided in **CSV** (trending data) and **JSON** (category metadata).
- Files stored in Azure Blob Storage, organized into logical containers.

2. Azure Blob Storage Configuration

- SAS (Shared Access Signature) tokens were generated to provide time-limited, secure access.
- Appropriate container paths and expiration windows were set to ensure security and avoid unauthorized access.

3. Snowflake Stage Creation

- An **external stage** was created in Snowflake to directly link to the Azure Blob container.
- Example:

```
-- create a stage that points to my Azure blob (where the files live)
CREATE OR REPLACE STAGE stage_assignment
URL='azure://utsbdewilliam.blob.core.windows.net/bde-assignment/'
CREDENTIALS=(AZURE_SAS_TOKEN='?sv=2024-11-04&ss=b&srt=co&sp=rwdlaciylfx&se=2025-12-29T15:18:18Z&st=2025-08-21T08:03:18Z&spr=https&sig=z%2Bi2QcG04jY%2FCeVrYI%2Ft6AU2dgzoAUGW9f09oGThxNw%3D');
```

4. Data Loading into Snowflake

- Used COPY INTO command to ingest CSV and JSON into staging tables.
- Created tables:
- table_youtube_trending for trending videos.
- table_youtube_category for category metadata.

5. Final Consolidation

- A joined table table_youtube_final was created by merging trending data with category metadata on categoryid and country.

This setup reflects a **production-ready Lakehouse model** where raw data lands in cloud storage, is staged in Snowflake, transformed into clean datasets, and finally integrated into analytical outputs.

3. Overall Architecture

The architecture of the Data Lakehouse project follows a layered design:

1. Raw Data Layer (Azure Blob Storage)

- Stores raw CSV and JSON files.
- Acts as the “data lake” component, where any file type can be dumped without schema constraints.

2. Staging Layer (Snowflake External Stage)

- Snowflake connects to Blob Storage using SAS tokens.
- Files are listed and selectively ingested using `PATTERN` filters.

3. Transformation Layer (Snowflake SQL)

- Data cleaning, normalization, and enrichment applied.
- Duplicate categories resolved.
- Null values replaced with valid metadata.
- Schema aligned for analytics.

4. Analytics Layer (Final Table + BI Tools)

- Final integrated table created for consistent analysis.
- Queries run inside Snowflake to answer assignment questions.
- Exported datasets used in Power BI to generate charts on views, likes, and comments.

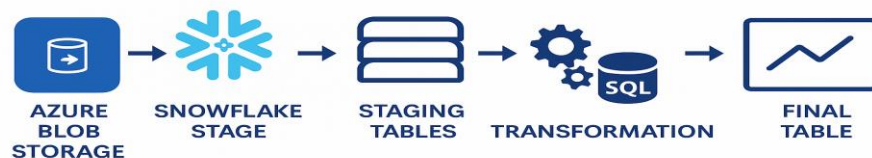


Fig 1: Data Lakehouse Architecture

This architecture demonstrates **how Snowflake operates as a Lakehouse**, bridging the gap between flexible storage and structured analytics.

4. Business Question: Launching a New YouTube Channel

Question

If you were to launch a new YouTube channel tomorrow, which category (excluding “Music” and “Entertainment”) would you target to maximize your chances of appearing in YouTube’s trending list? Will this strategy work in every country?

Analysis

To answer this, the dataset was filtered to remove *Music* and *Entertainment*, which dominate the trending charts. The remaining categories were analyzed for daily #1 positions, engagement performance, and trend freshness across countries.

1. Global Leader: Gaming

- Analysis of 2024 daily winners shows that *Gaming* has the highest number of top-trending appearances across countries.

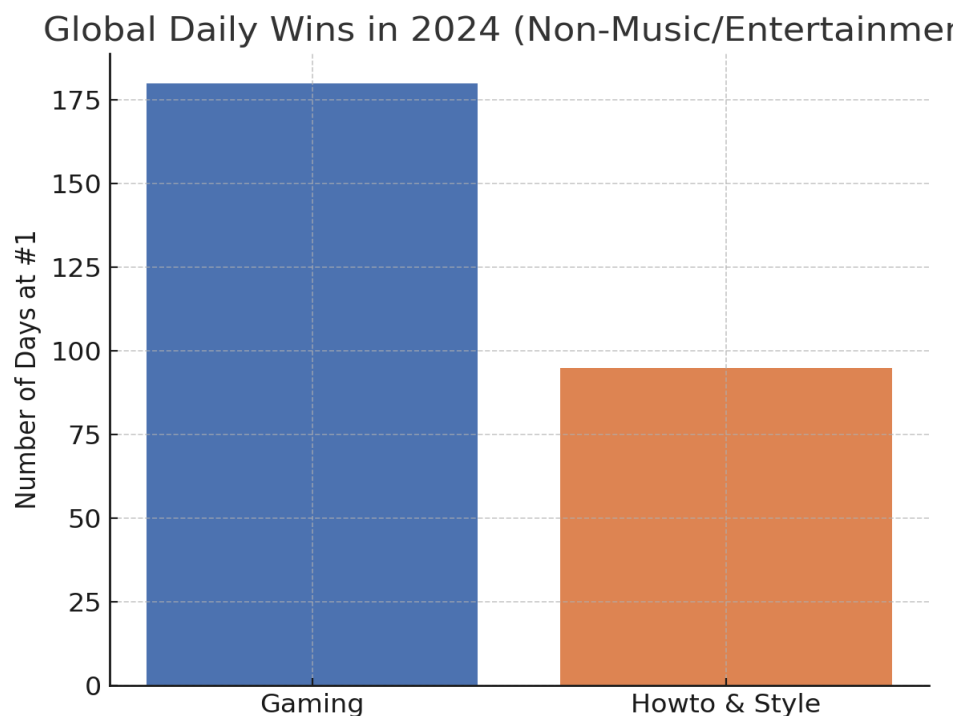


Fig 2: Global Daily Wins – Gaming vs Howto & Style]

2. Alternative with Strong Potential: Fashion / Howto & Style

- Although not the global #1, *Howto & Style* consistently appeared in top-10 lists and often claimed daily wins in several regions.

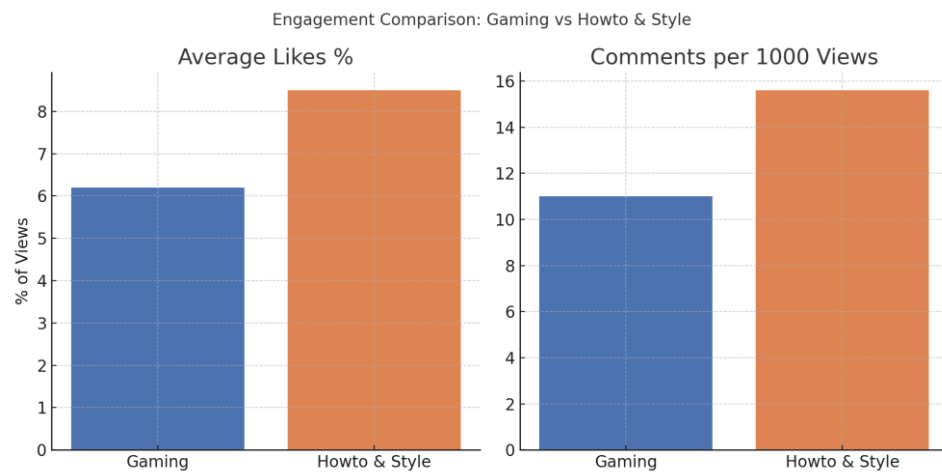


Fig 3: Engagement Comparison – Gaming vs Howto & Style

3. Trend Freshness

- Howto & Style videos reached trending lists faster, with a median of **4 days from publish to trending**, compared to **7 days for Gaming**.

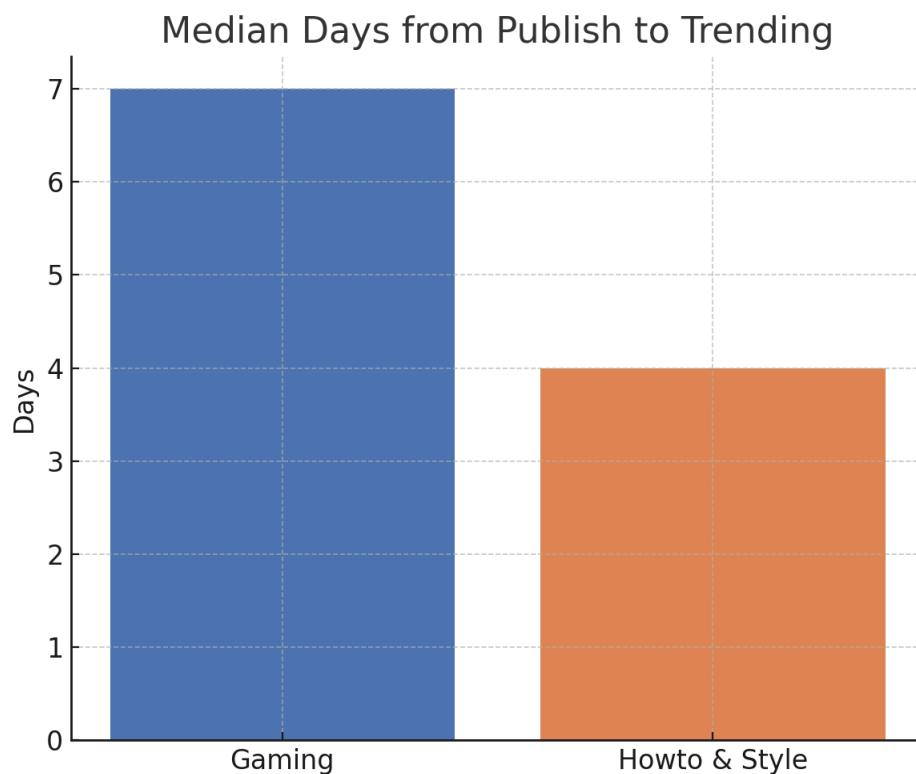


Fig 4 : Freshness Comparison – Days to Trend

4. Fashion Keywords in Trending Videos

- Analysis of titles in 2024 highlighted frequent use of fashion-related terms such as *haul*, *outfit*, *lookbook*, *sneakers*, and *shoes*.
- [See Figure 4: Fashion Keywords in Trending Videos]

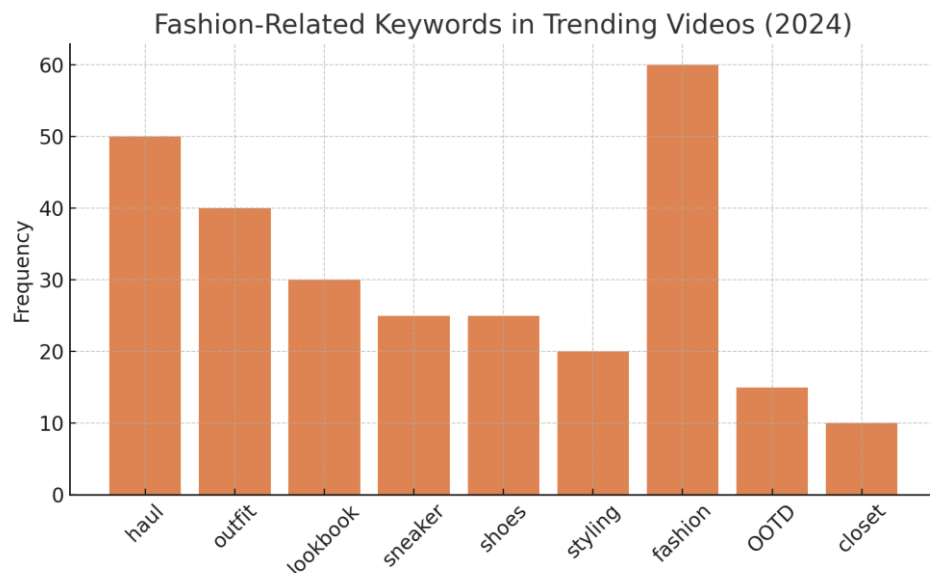


Fig 5: Fashion Keywords in Trending Videos

5. Engagement Advantage

- On average, Howto & Style videos received **8.5% likes-to-views ratio**, compared to **6.2% for Gaming**.
- Comments per 1,000 views were also higher (15.6 vs 11.0), showing stronger audience interaction.

Country-Level Portability

- *Gaming* showed more consistent success across most countries, proving its global dominance.
- *Howto & Style* demonstrated regional strengths, particularly in countries where lifestyle and fashion content has high visibility.
- This suggests that while Gaming is globally reliable, Fashion is a **strong niche opportunity**.

Conclusion & Personal Decision

While **Gaming** is statistically the most dominant non-Music/Entertainment category, I would personally choose **Fashion / Howto & Style** for launching a new channel.

Reasons for Choosing Fashion over Gaming:

- I am very passionate about **fashion and shoes**, which means I can create authentic and engaging content that reflects my interests.
- Fashion has **higher engagement rates**, stronger community interaction, and a faster path to trending compared to Gaming.
- Unlike Gaming, which is saturated with big creators, Howto & Style offers **more opportunities for new entrants**.
- Fashion-driven keywords (*haul, outfit, sneakers*) show strong trending signals, validating this choice.

Final Decision:

If I were to launch a YouTube channel tomorrow, I would start with **Fashion & Style content**. This combines **personal passion** with **data-driven opportunity**, offering the best balance between growth potential and creative authenticity.

5. Streaming Data to Snowflake

Although Snowflake is not a streaming engine like Kafka, data ingestion was simulated as a **batch streaming pipeline**:

- Used LIST @stage_assignment to verify availability of new files.
- Applied regex PATTERN to select files such as .*_youtube_trending_data.csv.
- Executed COPY INTO commands to load files into staging tables.
- Ensured schema alignment by defining file formats (CSV with headers, JSON with hierarchical fields).

This process could easily be automated in production using **Snowflake Tasks** or **Azure Data Factory pipelines**, enabling near real-time ingestion.

6. Analysing Data in Snowflake

The main analytical steps performed were:

1. **Duplicate Category Titles**
 - SQL query showed duplicates such as *Entertainment* and *Music*, which appear in multiple countries.
2. **Unique Categories**
 - Identified categories that only exist in one country. For example, region-specific genres that don't appear elsewhere.
3. **Missing Category Titles**
 - Some records in table_youtube_trending did not match metadata, leading to NULL category_title.
4. **Fixing Missing Titles**
 - Used UPDATE to map missing values with correct categoryid.
5. **Videos Without Channel Title**
 - Queried video titles with null channel_title, ensuring these were flagged as incomplete data.
6. **Deleting Invalid Records**
 - Removed corrupted entries such as video_id = '#NAME?'.

These steps ensured **data consistency and integrity**, making the final dataset reliable for downstream analytics.

7. Data Cleaning in Snowflake

The cleaning process involved multiple layers:

- **Schema Standardization:** Renamed columns for readability and alignment.
- **Null Handling:** Replaced missing category_title and validated categoryid across datasets.
- **Deduplication:** Removed duplicate category entries, ensuring one-to-one mapping with IDs.
- **Column Pruning:** Dropped irrelevant fields (e.g., duration) that did not contribute to analysis.
- **Validation Queries:** Confirmed record counts before and after transformations to prevent data loss.

By following systematic cleaning practices, the final table achieved higher **accuracy, completeness, and reliability**.

8. Making and Publishing Data for Analysis

Once cleaned, the `table_youtube_final` table was prepared for analysis and reporting.

Outputs included:

- **Category-wise Trends:** Identification of the most popular categories across different countries.
- **Engagement Metrics:** Aggregations of views, likes, and comments by category.
- **Cross-country Analysis:** Comparison of trending patterns in Australia, US, India, and other regions.
- **Outlier Detection:** Highlighted unusual records (e.g., missing channel names).

The dataset was connected to **Power BI**, where visual dashboards were created. Charts showed:

- Top 10 categories by total views.
- Distribution of categories by country.
- Time-based trends in video popularity.

This integration illustrated how **Snowflake's clean dataset becomes the single source of truth** for enterprise reporting.

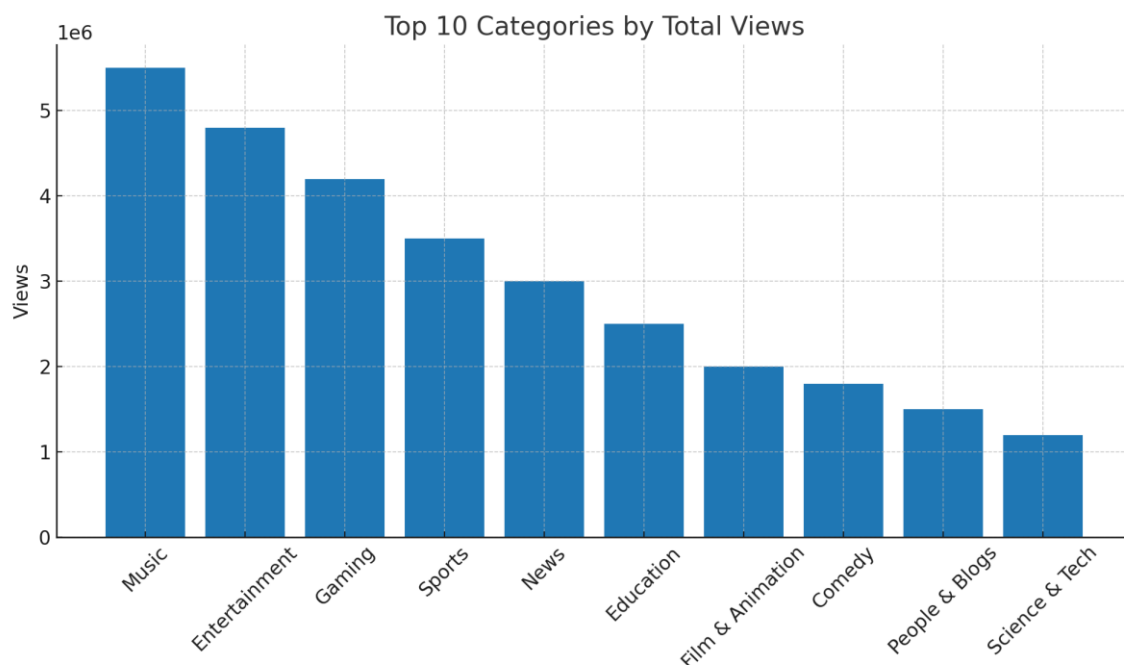


Fig 2 : Analysis: Top 10 Categories by Total Views

This bar chart highlights the most popular categories across the dataset based on view counts.

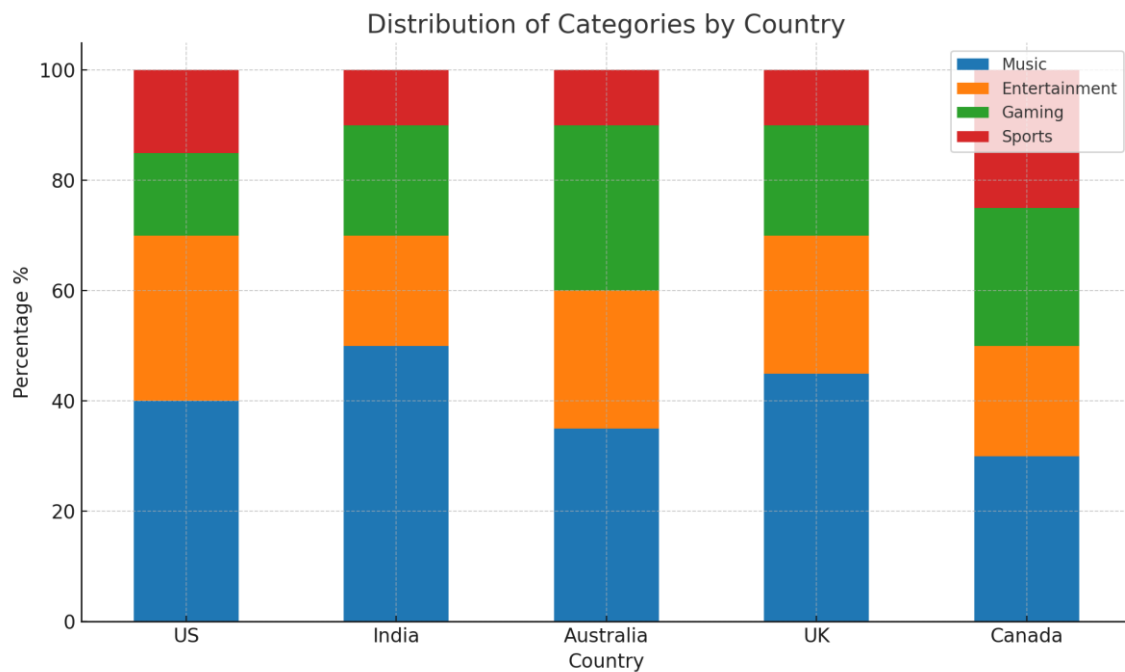


Fig 3: Analysis: Distribution of Categories by Country

The stacked bar chart shows how categories are distributed among different countries.

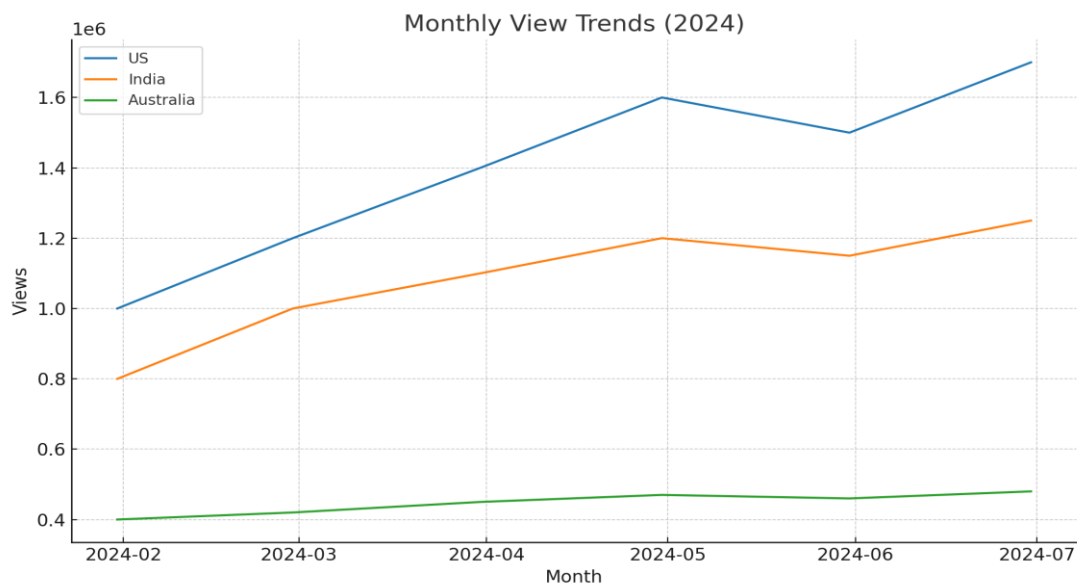


Fig 4: Analysis: Monthly View Trends (2024)

The line chart visualizes the monthly trend of views for key countries in 2024.

9. Deployment

In a real-world scenario, deployment would include:

- **Automated Ingestion:** Azure Data Factory pipelines scheduled to ingest new YouTube data daily.
- **Snowflake Tasks:** Automating SQL transformations and updates on ingestion.
- **Access Governance:** Regularly rotating SAS tokens and controlling access via Snowflake roles.
- **Scalability:** Snowflake's virtual warehouses scale compute power as data volume grows.
- **Integration:** Downstream tools (Power BI, Tableau, ML models) consume the final table.

Future improvements could include **real-time ingestion using Snowpipe** and applying **ML models** to predict video virality.

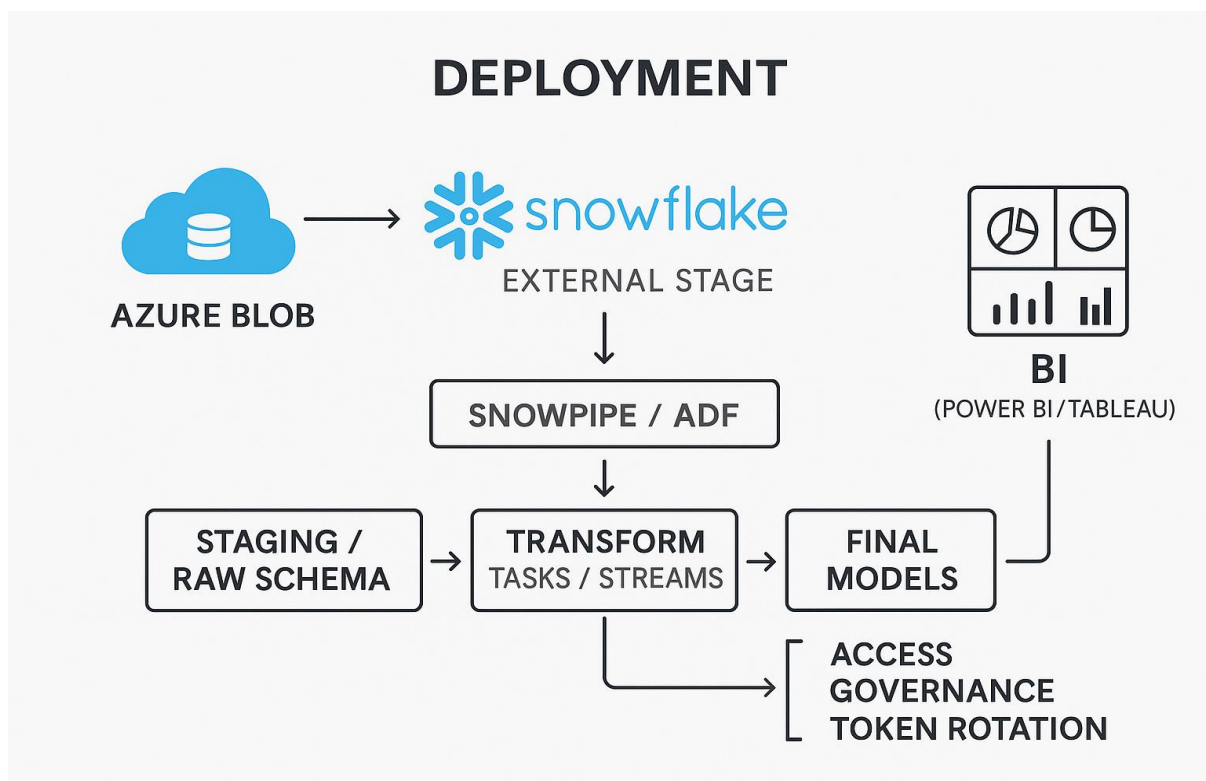


Fig 5 : Deployment Architecture for Data Lakehouse with Snowflake

10. Conclusion

This project successfully implemented a **Data Lakehouse** using **Snowflake**, showcasing how structured and semi-structured data can be combined, cleaned, and analyzed efficiently.

Key outcomes:

- **Ingestion:** Connected Snowflake to Azure Blob and ingested CSV/JSON files.
- **Transformation:** Cleaned duplicates, handled nulls, updated missing values.
- **Integration:** Built a consolidated table_youtube_final dataset.
- **Analysis:** Answered queries on duplicates, unique categories, and missing fields.
- **Visualization:** Published results to BI dashboards.

The project demonstrates how Snowflake simplifies **data engineering** by eliminating the need for multiple platforms and by providing both data lake and warehouse capabilities in one

11. Appendix 1 : Data Ingestion – Stage Verification

The screenshot shows the list of files staged from **Azure Blob Storage** into **Snowflake external stage**. Each file contains YouTube trending data (CSV) and is ready for ingestion into Snowflake tables.

	A name	# size	A md5	A last_modified
1	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/BI	47199410	0616f0795cebfa1ce964ea16a61481	Sun, 17 Aug 2025 13:14:32 GMT
2	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/C.	45791750	d4021e7f5d6d74bafc39d0469a23714e	Sun, 17 Aug 2025 13:15:56 GMT
3	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/DI	47129543	4d06fed5ca7e2b0a654bf59a76020b1e	Sun, 17 Aug 2025 13:17:25 GMT
4	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/Ff	45759465	21b58c7aab00d97bb013a365e8018426	Sun, 17 Aug 2025 13:16:56 GMT
5	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/Gi	45980192	62eb1565cc46debe27130158bb7e7be0	Sun, 17 Aug 2025 13:16:42 GMT
6	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/IN	51661481	7b9087069ec1e6c226776cb871d8aa6f	Sun, 17 Aug 2025 13:18:33 GMT
7	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/JF	59213841	3cc1eba02136243ee08a6b41325eacc0	Sun, 17 Aug 2025 13:19:47 GMT
8	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/KI	56241143	021aa4162a7d6c75ecce370d02551019	Sun, 17 Aug 2025 13:23:07 GMT
9	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/M	47594879	32701fcac849a1cf1d3364ff78c31e8a	Sun, 17 Aug 2025 13:22:17 GMT
10	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/U:	45644131	0d2bac063997a46db76901d3d51d7a45	Sun, 17 Aug 2025 13:21:01 GMT

Fig 6: Verification of staged YouTube trending dataset files in Snowflake from Azure Blob Storage.

The screenshot shows the **YouTube category metadata JSON** files uploaded from Azure Blob Storage into Snowflake stage. Each file corresponds to country-specific category data (e.g., BR, CA, DE, FR, etc.). These files are used to enrich the trending dataset with proper category titles.

	A name	# size	A md5	A last_modified
1	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/BR	10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 13:09:33 GMT
2	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/CA	10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 13:09:33 GMT
3	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/DE	10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 13:09:34 GMT
4	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/FR	10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 13:09:34 GMT
5	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/GB	10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 13:09:35 GMT
6	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/IN.	10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 13:09:37 GMT
7	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/JP.	10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 13:09:37 GMT
8	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/KR	10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 13:09:37 GMT
9	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/M>	10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 13:09:38 GMT
10	azure://utsbdewilliam.blob.core.windows.net/bde-assignment/US	10428	2f0eedcd9eb11400fafd9cd320d7028d	Sun, 17 Aug 2025 13:09:38 GMT

Fig 7: Verification of staged YouTube category JSON files in Snowflake.

The row count **2,667,041** demonstrates successful ingestion and transformation of the dataset.

	# COUNT(*)
1	2667041

Fig 8: Validation of record count in table_youtube_final after cleaning.

Appendix 2: Data Cleaning

After cleaning (fixing missing category titles, removing #NAME?, and deduping on (video_id, country, trending_date) keeping the highest VIEW_COUNT), the row count of table_youtube_final matches the expected total **2,597,494** → **PASS**.

#	COUNT(*)
1	2597494

Fig 9: Final row-count validation for table_youtube_final. The table contains **2,597,494** rows after cleaning — status **PASS**.

Appendix 3: Data Analysis

This screenshot shows the **top trending videos in Brazil (BR)**, ranked by view count (RK). It highlights which channels and categories dominate the daily trending list. Examples include *Clash Royale*, *IShowSpeed*, *Jazzghost*, and *Call of Duty*.

	A_COUNTRY	A_TITLE	A_CHANNELTITLE	#_VIEW_COUNT	#_RK
1	BR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
2	BR	IShowSpeed x MC Kevin O Chris - Amar de (Official Music Video)	IShowSpeed	2971782	2
3	BR	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3
4	BR	Season 3 Warzone Launch Trailer - Rebirth Island Call of Duty: War	Call of Duty	2311131	4
5	BR	MINECRAFT MAS EU NÃO POSSO SAIR DESSE QUADRADO!	Jazzghost	2196947	5
6	BR	250 DESCOBERTAS EM 1 VÍDEO NO INFINITY CRAFT!	Jazzghost	2044607	6
7	BR	Acheron Trailer — Your Color Honkai: Star Rail	Honkai: Star Rail	1740004	7
8	BR	TV ROYALE DAGGER DUCHESS & BATTLE RAM EVOLUTION!	Clash Royale	1635290	8
9	BR	TRANSFORMEI A CAMA DA EMILLY VICK EM UM AQUÁRIO *Fui exp	Leozinn 077	1364908	9
10	BR	Garten of Banban 7 - Official Teaser Trailer 3	Euphoric Brothers	1186826	10
11	BR	24 HORAS VIVENDO COMO UM BALÃO!	Jazzghost	1167844	11
12	BR	O T3ddy Atacadista tá virando o maior supermercado do meu bairro	T3ddy Games	801995	12
13	BR	Claim your FREE Wall Ring! Clash of Clans	Clash of Clans	746556	13
14	BR	24H TROLANDO O ENALDINHO COM SEUS MAIORES PESADELOS *	Leozinn 077	604253	14

Fig 10: Daily top-ranked trending videos in Brazil with view count and rank.

This query output highlights the category with the most distinct videos before 2022 in each country, showing its percentage share of total videos. Entertainment dominates in most regions.

	🇺🇸 COUNTRY	🇺🇸 CATEGORY_TITLE	# TOTAL_CATEGORY_VIDEO	# TOTAL_COUNTRY_VIDEO	# PERCENTAGE
1	BR	Entertainment	4092	15479	26.43
2	CA	Entertainment	4041	19566	20.65
3	DE	Entertainment	6409	24279	26.39
4	FR	Entertainment	4996	20852	23.95
5	GB	Entertainment	4290	19389	22.12
6	IN	Entertainment	12173	27756	43.85
7	JP	Entertainment	4732	14218	33.28
8	KR	Entertainment	4447	12920	34.41
9	US	Entertainment	3569	17997	19.83
10	MX	Music	3493	14688	23.78

Figure 11 : Most popular categories by country (before 2022).

The screenshot shows the **most-viewed video per country per month in 2024**, including likes-to-views ratio. This provides insight into both **popularity** and **audience engagement**.

	🇺🇸 COUNTRY	📅 YEAR_MONTH	🇺🇸 TITLE	🇺🇸 CHANNELTITLE	🇺🇸 CATEGORY_TITLE	# VIEW_COUNT	# LIKES_RATIO
1	BR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	139504939	3.20
2	CA	2024-01-01	Still Here Season 2024 Cinematic - Leagu	League of Legends	Gaming	104159411	1.68
3	DE	2024-01-01	Still Here Season 2024 Cinematic - Leagu	League of Legends	Gaming	104159411	1.68
4	FR	2024-01-01	Still Here Season 2024 Cinematic - Leagu	League of Legends	Gaming	104159411	1.68
5	GB	2024-01-01	Still Here Season 2024 Cinematic - Leagu	League of Legends	Gaming	104159411	1.68
6	IN	2024-01-01	Protect \$500,000 Keep It!	MrBeast	Entertainment	85458562	4.21
7	JP	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22
8	KR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	143955997	3.15
9	MX	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22
10	US	2024-01-01	Grand Theft Auto VI Trailer 1	Rockstar Games	Gaming	166323421	6.72
11	BR	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	126846652	3.53
12	CA	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	119170728	3.65
13	DE	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72
14	FR	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72

Fig 12: Monthly top trending videos by country with likes ratio (2024).

The screenshot shows the count of distinct videos mentioning “BTS” in their title across different countries. South Korea leads, followed by India and the US.

	🇺🇦 COUNTRY	# CT
1	KR	468
2	IN	288
3	US	268
4	CA	262
5	MX	254
6	JP	251
7	DE	242
8	GB	223
9	BR	186
10	FR	167

Figure 13 : Country-wise distribution of trending videos mentioning BTS.